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QAOA-based MRMR Algorithm for Feature Selection

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ABSTRACT

As the volume of data for classification tasks in machine learning grows, feature selection plays an increasingly crucial role in enhancing the efficiency and effectiveness of these tasks. Existing classical feature selection algorithms often encounter challenges, such as high computational complexity and vulnerability to local optima. Quantum algorithms have been proposed to overcome these limitations. Here, our work proposes a new approach that combines the Quantum Approximate Optimization Algorithm (QAOA) with the classical feature selection algorithm Maximum Relevance-Minimum Redundancy (MRMR). First, we transform the feature selection problem into the Quadratic Unconstrained Binary Optimization (QUBO) problem. Then, we map the QUBO formulation to Hamiltonian with the Ising model. Finally, we employ QAOA with a shallow circuit to search this Hamiltonian's ground state, which matches the specific solution for feature selection. We conducted numerical experiments on eight real-world datasets for classification tasks from diverse domains, including finance, medicine, and natural hazards. Experiment results suggest that QAOA-selected feature subsets size are typically less than half the complete set while achieving comparable or better classification accuracy than classical MRMR, demonstrating its preference for smaller, more efficient feature subsets.

CCS CONCEPTS

• **Computing methodologies** → Machine learning; Machine learning algorithms; Feature selection.

KEYWORDS

QAOA, Feature selection, Classification

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1 INTRODUCTION

Nowadays, the rapid growth and advancement of machine learning have resulted in its widespread application across various domains. Consequently, the volume of data used to train machine learning models has experienced a remarkable exponential growth [1]. In classification tasks, this overwhelming deluge of data often contains redundant information, which hampers classifier efficiency and performance. To reduce the dimensionality of the dataset, the papers propose various classical feature selection methods [2], [3], [4], [5], [6]. These approaches are valuable but constrained by computational power and storage complexity. Some methods, such as Maximum Relevance-Minimum Redundancy (MRMR) Algorithm [7], aim to identify the subset of features that exhibit the highest level of information and relevance in a given dataset [8]. These methods provide the advantage of lower computational complexity. Nonetheless, their susceptibility to getting trapped in local optima is a notable defect.

Quantum algorithms have been proposed to address the local optima dilemma [9], [10], [11], [12], [13]. One such algorithm is the Quantum Approximate Optimization Algorithm (QAOA) [9], a variational quantum algorithm that combines classical optimization with the quantum circuit. QAOA is capable of finding approximate solutions to optimization problems, even those with local optima.

Many Quantum Machine Learning (QML) works have already investigated the combination of quantum advantage with classical algorithms to overcome their inherent limitations [14], [15], [16]. However, the error rates of these initial machines significantly constrain the total number of gates that can be executed reliably, and those quantum algorithms with promised results are beyond our reach before [17]. For instance, in the study conducted by Turati et al. [16], their QAOA may require the implementation of p=10 layers of gates to achieve superior results when executed on a 7-qubit IBM quantum computer. The quantum circuit burden of these algorithms can be relatively high, which keeps them from being well implemented on existing devices. Therefore, algorithms, such as QAOA and VQE that could fit in the NISQ devices will be

useful tools for exploring meaningful applications. Overall, finding a more efficient and scalable feature selection method is crucial for quantum machine learning research.

This work focuses on integrating the QAOA with the classical feature selection algorithm MRMR. We convert the MRMR feature selection problem into the Quadratic Unconstrained Binary Optimization (QUBO) problem [18]. To solve the issue of division by zero in the Hamiltonian encoding, we applied polynomial fitting. Subsequently, we employed the Ising model for the encoding process. By leveraging QAOA with p=2 layers for ground state search, we achieve approximate global solutions significantly superior to MRMR searches, demonstrating the potential for efficient use of near-term quantum devices. In summary, the main contributions of the article are illustrated as follows:

- The feature selection algorithm QAOA-based MRMR demonstrates a superior or comparable classifier score compared to the classical MRMR algorithm when subjected to the same threshold of feature size.
- The algorithm exhibits a greater inclination to select a small feature subset with a high probability while achieving a high classification score.
- The implementation of high-precision polynomial fitting offers a concise and effective solution to encountering a denominator of 0 in the Hamiltonian encoding of the cost function within the quantum circuit.

2 METHODS

2.1 Preliminaries

2.1.1 Maximum Relevance-Minimum Redundancy (MRMR) Algorithm. The MRMR algorithm is a classical feature selection technique that aims to identify the most representative features from the complete feature set. By selecting the optimal subset of features for modeling, the algorithm can effectively preserve the correlation information between target indicators and features while minimizing the redundancy among the selected features. This algorithm was designed based on the following principle: given two random variables X, Y their probability density function corresponding to the $p(x)$, $p(y)$, then the mutual information I(x, y) is defined as:

$$\mathrm{I}(\mathrm{x}:\mathrm{y}) = \iint p(x,y) \log \frac{p(x,y)}{p(x)\,p(y)} dxdy \tag{1}$$

The correlation between feature set S and class C is defined as the average of the mutual information between each feature f_i and class C.

$$\mathrm{D}(\mathrm{S},\mathrm{C}) = \frac{1}{S} \sum_{f_i \in S} I(f_i; C) \tag{2}$$

The redundancy of all features in the set S is defined as the average of the mutual information between the features f_i and f_j.

$$\mathrm{R}(\mathrm{S}) = \frac{1}{S^2} \sum_{f_i, f_j \in S} I(f_i; f_j) \tag{3}$$

The cost function aims to find the set S with the maximum correlation and the minimum redundancy, and directly maximize the following function $\Phi = D - R$:

$$\Phi(\mathrm{D},\mathrm{R}) = \frac{1}{S} \sum_{f_i \in S} I(f_i; C) - \frac{1}{S^2} \sum_{f_i, f_j \in S} I(f_i; f_j) \tag{4}$$

The defect of this algorithm is obvious: it is a non-global optimal solution algorithm. Based on the empirical test results, there is still a notable discrepancy between the obtained solutions and the optimal solution when we use MRMR. If the traversal algorithm of the global optimal solution were used, the time complexity required by the algorithm is probably exponential.

Quadratic Unconstrained Binary Optimization (QUBO). The maximization problem presented in Equation 4 can be converted into a QUBO problem [18]:

$$\min_{x \in \{0,1\}^n} x^T Q x \tag{5}$$

where x = (x1, ..., xn) $\in \{0, 1\}^n$ and Q is real-valued upper triangular matrix, whose entries Q_{ij} define a weight for each pair of indices i, j $\in \{1, \ldots, n\}$ within the binary vector [19].

2.1.2 Quantum Approximate Optimization Algorithm (QAOA).. The QAOA provides an approximate solution to QUBO problems. What sets it apart from other variational algorithms is that the quantum circuit is defined by the specific problem instance, featuring very few adjustable parameters. Additionally, it is widely believed that circuits with lower depth are sufficient for obtaining high-quality approximate solutions [20].

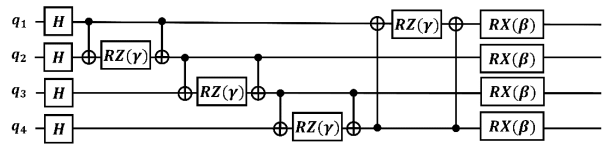

Figure 1: Overview of the QAOA.

2.2 MRMR Based on QAOA

2.2.1 Hamiltonian Encoding. The classical computer operates using binary numbers 0 and 1, whereas the quantum computer relies on the Qubit $|Z\rangle$. Within the framework of the QAOA, the algorithm is designed to encode the correlation between the objective function and the binary vector into the Hamiltonian, so that a certain energy eigenstate corresponds to a certain binary vector, and its energy corresponds to the objective function [11], [21].

In the MRMR algorithm, the Hamiltonian corresponding to the problem is the sum of all the Hamiltonian which corresponds to a feature selection way, and this Hamiltonian is of the cost layer (HC) in QAOA. We optimize the parameters to make the cost function as large as possible, obtaining the parameter pair (γ, β). Subsequently, we construct the corresponding state $|Z\rangle$ through (γ, β), representing the selected features for the solution of the problem.

However, the denominator in the loss function of MRMR, which are $\frac{1}{S}$ and $\frac{1}{S^2}$, can potentially become zero given by Equation 4. This poses a significant challenge when encoding the problem into the Hamiltonian framework. To overcome this obstacle, we utilize a polynomial curve approach based on the method of least squares to

approximate the terms $\frac{1}{S}$ and $\frac{1}{S^2}$. By employing second and third-order polynomials to approximate the terms $\frac{1}{S}$ and $\frac{1}{S^2}$, respectively, we effectively mitigate the issue and allow for the successful encoding of the problem into the Hamiltonian framework.

2.2.2 Operators and Implementation Feasibility. We identify the mapping between the MRMR cost function and the basis for diagonal operators. We employ the Ising model to represent the problem solution within the ground state of HC. A classical Ising model can be expressed as a quadratic function involving a set of n spins denoted as $s_i = 1$ [22]. The quantum version of its Hamiltonian is straightforward:

$$H_P = H\left(\sigma_1^z, \ldots, \sigma_n^z\right) \tag{6}$$

where σ_i^z represents a Pauli matrix (a 2×2 matrix), and its cousin $\frac{(1+\sigma_i^z)}{2}$ has eigenvectors $|0, 1\rangle$ with corresponding eigenvalues 0 and 1. This matrix operates on the ith qubit within a Hilbert space comprising n qubits $\{|+>, |->\}^{\otimes n}$. Thus, as long as the corresponding relationship is taken: $z_i \longrightarrow \frac{1-z_i}{2}$, the quantum state can correspond to binary in the computational basis $|0\rangle \longrightarrow 0$, $|1\rangle \longrightarrow 1$, and the Hamiltonian corresponding to the objective function is as follows:

$$H_C = C\left(\frac{I - Z_1}{2}, \frac{I - Z_2}{2}, \ldots, \frac{I - Z_{n-1}}{2}, \frac{I - Z_n}{2}\right) \tag{7}$$

Therefore any diagonal unitary may be written as a product of exponentials of operators:

$$\mathrm{U} = e^{iHt} = \prod_{j=0}^{n-1} e^{ia_i a_z^j} \tag{8}$$

Each term in the product takes the form of $e^{-i\frac{\theta_j}{2}\otimes_i (Z_i)^j}$, with $\theta_j = -2a_j$. Hence, the circuit for the Z-gate is replaced by a Z-rotation, $R_z(-2a_j)$, where $R_Z(\theta) = e^{-iZ\frac{\theta}{2}}$. The QAOA circuit with 4 qubits, $p = 1$ will look like Figure 1. The implementation is subject to the limitation of gate fidelity. In noisy quantum computers, increasing the number of layers can lead to even worse accuracy due to gate errors negating the benefits of higher [9]. In our work, we use $p = 2$ layers, creating a shallow quantum circuit with low depth, which suffices to achieve good approximate solutions [23], [24].

3 EXPERIMENTS AND RESULTS

We perform simulated experiments with two goals:

1. To compare QAOA-based MRMR with classical MRMR for feature selection in classification models.
2. To quantify the probability of QAOA discovering the ground state.

3.1 Select Feature with QAOA

The primary aim in searching for an optimal feature subset is to identify a set of features capable of effectively enhancing the classification accuracy in the analysis of the original, high-dimensional dataset. To accomplish this objective, a secondary goal is to simplify the complex high-dimensional dataset into a lower-dimensional form. In our work, we analyze eight real-world datasets derived from diverse domains [25], encompassing natural hazards, finance,

Table 1: Polynomials curve metrics.

Dataset		Curve Metrics	
Name	Feature Size	MSE	R2
Breast Cancer (Wisconsin)	9	4.559e-5	0.9528
Heart Failure Clinical Records	12	1.825e-5	0.9619
Algerian Forest Fire	12	3e-4	0.8356
Australian Credit Approval	14	1.745e-5	0.9618
Audit Data	17	5.842e-5	0.9291
Cervical Cancer Behavior Risk	19	2.074e-5	0.8618
South German Credit	20	1.708e-6	0.9001
Glioma Grading Clinical	23	2.831e-6	0.9845

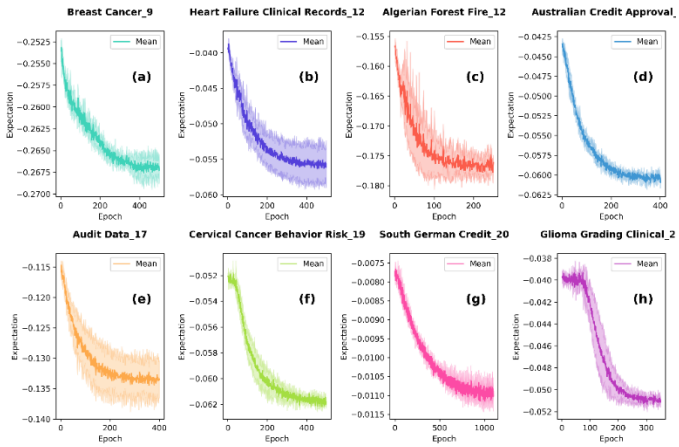

Figure 2: The output expectation of QAOA.

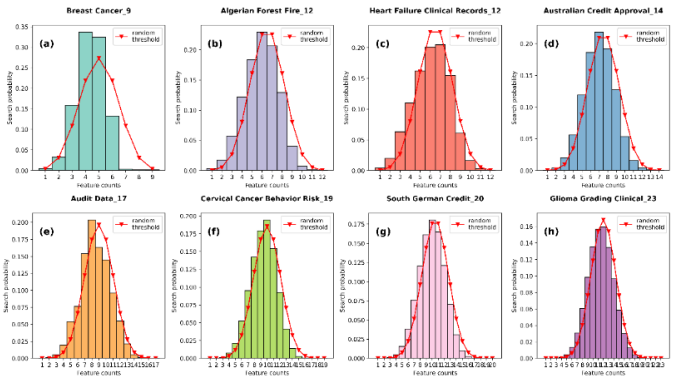

Figure 3: The probabilities of QAOA and random search for different feature subsets.

Table 2: Results of the classifier for test datasets of various feature sizes.

Dataset	Breast Cancer		Heart Failure Clinical Records		Algerian Forest Fire		Australian Credit Approval	
Size	MRMR	QMRMR	MRMR	QMRMR	MRMR	QMRMR	MRMR	QMRMR
1	**0.928259**	0.923949	0.962705	**0.962705**	0.826081	**0.826081**	0.854970	**0.854970**
2	**0.959819**	0.951199	0.942418	**0.966667**	0.822838	**0.842793**	0.620303	**0.857945**
3	**0.956970**	0.956921	0.896585	**0.954577**	0.840036	**0.846036**	0.615926	**0.852189**
4	0.956962	**0.962693**	0.954781	**0.967008**	0.685631	**0.822613**	0.655044	**0.853609**
5	0.961273	**0.964130**	0.958811	**0.950546**	0.678919	**0.852973**	0.658052	**0.855046**
6	0.965550	**0.965591**	**0.950410**	0.954781	0.678919	**0.835991**	0.656489	**0.772458**
7	0.961281	**0.969885**	0.913320	**0.954508**	0.692252	**0.842928**	0.659371	**0.840587**
8	0.966987	**0.968407**	0.904645	**0.966872**	0.682252	**0.685631**	0.655070	**0.815978**
9	0.969893	**0.969893**	0.909016	**0.962773**	0.678919	**0.836126**	0.652221	**0.815978**
10			0.904713	**0.946175**	0.678919	**0.685586**	0.655002	**0.711604**
11			0.896721	**0.958743**	0.678919	**0.678919**	0.658069	**0.658069**
12			0.896585	**0.896585**	0.678919	**0.678919**	0.650760	**0.659514**
13							0.650743	**0.657901**
14							0.653633	**0.653633**
Optimal Size	9	**9**	1	**4**	3	**5**	1	**2**
Max Score	0.969893	**0.969893**	0.962705	**0.967008**	0.840036	**0.852973**	0.854970	**0.8579452**
Dataset	Audit Data		Cervical Cancer Behavior Risk		South German Credit		Glioma Grading Clinical	
Size	MRMR	QMRMR	MRMR	QMRMR	MRMR	QMRMR	MRMR	QMRMR
1	0.628719	**1.000000**	0.644608	**0.718137**	0.689000	**0.700000**	**0.846081**	0.576390
2	0.628719	**0.769704**	**0.901144**	0.790033	0.708000	**0.708000**	**0.853252**	0.733983
3	0.628719	**1.000000**	0.843954	**0.915033**	0.706000	**0.715000**	**0.739810**	0.738710
4	0.628719	**0.776174**	0.887255	**0.915033**	0.707000	**0.716000**	0.741051	**0.846058**
5	0.737327	**0.778825**	0.900327	**0.944444**	0.707000	**0.717000**	0.742196	**0.742196**
6	0.753025	**0.769724**	0.901961	**0.929739**	0.701000	**0.731000**	0.743438	**0.867538**
7	0.758146	**0.769684**	0.901144	**0.944444**	0.704000	**0.737000**	0.739810	**0.869936**
8	0.714091	**0.770999**	0.915033	**0.915850**	0.712000	**0.748000**	0.741091	**0.866359**
9	0.751636	**0.778744**	0.915850	**0.930556**	0.703000	**0.748000**	0.737440	**0.868763**
10	0.751683	**0.864243**	0.900327	**0.943627**	0.704000	**0.742000**	0.737497	**0.869908**
11	0.728320	**0.773623**	0.929739	**0.943627**	0.712000	**0.747000**	0.742253	**0.872289**
12	0.747743	**0.751596**	0.929739	**0.944444**	0.709000	**0.742000**	**0.749356**	0.742265
13	0.754233	**0.761932**	0.901144	**0.943627**	0.712000	**0.735000**	0.735059	**0.871098**
14	0.749072	**0.751696**	0.915850	**0.929739**	0.710000	**0.712000**	0.739833	**0.869891**
15	0.745112	**0.750394**	0.929739	**0.943627**	0.708000	**0.721000**	0.738625	**0.867527**
16	0.751589	**0.752978**	0.914216	**0.915850**	0.709000	**0.710000**	0.743404	**0.868763**
17	0.746381	**0.746381**	0.914216	**0.928922**	0.709000	**0.712000**	0.738716	**0.871110**
18			0.901144	**0.915850**	0.704000	**0.709000**	0.741051	**0.867567**
19			0.901144	**0.901144**	0.706000	**0.712000**	0.742259	**0.782855**
20					0.706000	**0.706000**	0.742225	**0.863944**
21							0.739878	**0.743484**
22							**0.744612**	0.743524
23							0.741029	**0.741029**
Optimal Size	7	**1**	11	**5**	8	**8**	2	**11**
Max Score	0.758146	**1.000000**	0.929739	**0.944444**	0.712000	**0.748000**	0.853252	**0.872289**

and the medical field. The QAOA circuit comprises $p = 2$ layers, in which gates representing the cost operator $U_C(\gamma)$ and the mixer operator $U_M(\beta)$ are applied [9]. The datasets have different feature sizes, specifically 9, 12, 12, 14, 17, 19, 20, and 23 features for each dataset. These datasets are specifically curated for classification tasks, serving as benchmarks to evaluate the performance and efficacy of our algorithm with a Support Vector Machine (SVM) classifier using 5-fold cross-validation. We employed the QAOA for ground state search, using a fixed number of 1000 sampling shots. The optimizer ADAM was utilized [26], and the convergence of iterations exhibited variations, ranging from 150 to 800 generations, across each dataset, as illustrated in Figure 2.

The results of the classifier were displayed in Table 1. Comprehensive experiments conducted on eight publicly available datasets demonstrate that our approach consistently delivers exceptional performance while effectively reducing dimensionality. It is apparent that across the majority of feature scales, the QAOA-based MRMR consistently outperforms classical MRMR by achieving higher classification accuracy. Taking the Cervical Cancer Behavior Risk dataset as an example, QAOA selects the optimal feature size of 5, achieving a classifier accuracy of 94.4444%, while MRMR chooses an optimal feature size of 11, reaching a classifier accuracy of only 92.9739%. Furthermore, if we aim to achieve the optimal performance of MRMR by QAOA-based MRMR, we can pick a feature size of 6. It is worth noting that our proposed algorithm uses fewer features to make higher classification accuracy compared to the MRMR algorithm. All the presented classifier results unequivocally demonstrate the superiority of our work.

3.2 Impact of $\frac{1}{S}$ curve Fitting on Ground State Search

By applying a polynomial function fit, we successfully address the challenge of encoding the Hamiltonian with a zero denominator. Consequently, it becomes imperative to quantitatively assess the influence of polynomial curve-fitting on the

MRMR loss function and explore its implications for the QAOA ground state search. The loss function is evaluated and calculated for all cases, and the fitting approximation of the total loss function resulting from the approximation of $\frac{1}{S}$ and $\frac{1}{S^2}$ is assessed using metrics, including mean squared error (*MSE*) and R-squared (R^2) score. These metrics, which are utilized as numerical measures to assess the accuracy of the fitting approximation, are presented in Table 2. All MSE scores approach zero, while the R^2 scores exceed 0.8, indicating a highly accurate fit between the predicted and actual values. In other words, the encoded approximate loss function in the Hamiltonian closely approximates the original function, resulting in minimal disruption to the ground state search.

3.3 Probability of Ground State Search

Figure 3 depicts the probability of obtaining an acceptable set of features upon QAOA reaching iteration convergence. The red downward triangle line represents the probability of random feature selection in different feature sizes, and the histogram shows the probability of QAOA. Upon analysis, we can observe that, in contrast to the random feature selection approach, the results indicate a clear tendency of the QAOA to favor a smaller number of features during the selection process. Specifically, we find that the size of the feature subset selected by QAOA is primarily around half or less of the complete set.

4 CONCLUSION AND FUTURE WORKS

Our work demonstrates the practicality of employing the QAOA in conjunction with the classical algorithm MRMR for feature selection. The results in various classification tasks by QAOA-based MRMR all outperform those achieved through the classical MRMR feature selection method. Furthermore, our findings indicate that the QAOA exhibits a preference for selecting a smaller subset of features with a high probability. Additionally, the utilization of high-precision polynomial fitting presents a succinct and effective resolution to address the issue of a denominator of 0 in the cost function for Hamiltonian encoding.

The results presented were obtained on a simulated quantum computer, and the forthcoming manuscript will provide an in-depth account of the expansion of this research into real hardware implementations. In the future, we hope to extend the application of QAOA to other classical feature selection algorithms. The combination of these techniques contributes to the advancement of feature selection in the realm of both quantum computing and machine learning.

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